Emergent Fuzzy Rule-Like Computation in Transformers: Functors, Learned Categories, and State-Conditioned Residual Rewriting

Working Paper 0.7

Preregistered protocol and dependency audit — August 2026

Abstract

We preregister a controlled study of whether text-completion transformers develop computation usefully described as fuzzy, quasi-symbolic rule application. The study depends on both causally validated predictive equivalence (Paper 0.5) and behaviorally competent learned semantic classes (Paper 0.6). A machine-readable dependency audit finds that Paper 0.5 satisfies its prerequisite, but zero of four Paper 0.6 runs passes the fixed 80% held-out completion gate; observed accuracies range from 1.4% to 15.3%. Paper 0.7 is therefore blocked before semantic type-dispatch, composition, or rule-learning experiments. We retain the complete factorial design, entropy ontology, interventions, controls, and falsification criteria so that a future competent semantic replication can activate the protocol without changing its hypotheses after seeing results. The present result is a disciplined stop decision, not evidence for or against fuzzy rule-like computation.

1 Motivation

N-grams provide a primitive lexical mapping from prefix to continuation. Semantic learning adds equivalence classes and hierarchical relations. Their combination suggests transformations

$$F(x) \to y,$$

where a syntactic/relational frame F operates on an argument x, while the continuation depends on both frame and learned semantic type. This resembles a fuzzy object-oriented/frame-logic organization: soft entities, soft types, approximate matching, and reusable transformations. We do not claim a literal symbolic interpreter.

2 Research Questions

RQ1. Are syntactic/relational functors represented with substitution-tolerant invariants?

RQ2. Does learned category membership constrain the transformation induced by a functor?

RQ3. Does behavior generalize to unseen members and unseen member–functor combinations?

RQ4. Are transformations compositional across nested/chained functors?

RQ5. How do attention and MLP sublayers divide binding, category activation, and transformation?

RQ6. How does predictive entropy change while rule-consistent candidates are selected?

RQ7. Do causal interventions support the rule-like interpretation?

3 Hypotheses

H1: Soft functor invariance. Equivalent frames share a recoverable transformation component across argument substitutions.

H2: Learned type-sensitive dispatch. The same frame can induce different continuation/transformation families for different learned semantic categories.

H3: Category generalization. Learned behavior transfers to held-out category members.

H4: State-conditioned rewriting.

$$r_{l+1} = r_l + \Delta_l, \qquad \Delta_l = F_l(r_l).$$

Layers are therefore complementary causal stages, not independent estimators.

H5: Uncertainty restructuring. Successful inference tends to concentrate probability on rule-consistent equivalence classes over depth, although individual layers may expand or redirect the candidate set.

4 Training and Competence Gate

Use a small fully instrumentable causal transformer, with a pretrained small LM as an external-validity replication where practical. The custom model must first demonstrate held-out text-completion competence. Synthetic category labels are evaluation metadata only and must not be injected as privileged supervision in the main condition. Evaluate undertrained, threshold, and converged checkpoints. Residual category claims made before behavioral competence are invalid for the main hypothesis.

4.1 Current dependency-audit result

The audit consumes the committed Paper 0.5 v2 manifest and Paper 0.6 competence table. Predictive equivalence is available from Paper 0.5: equivalent-donor residual replacements preserve output distributions far better than nonequivalent replacements. Paper 0.6, however, has no competent run. Its four held-out accuracies are 1.4%, 4.9%, 4.9%, and 15.3%, all below the 80% threshold. Consequently no Paper 0.6 representation, centroid, hierarchy label, or common residual direction is admissible as a learned semantic type. E1–E6 are not run, and no result-dependent redesign is permitted before a competent replication.

5 Dataset

Construct a controlled textual world containing entities whose overlapping observable properties make category and hierarchy structure statistically learnable through ordinary completion. Include instance–subcategory–category–supercategory relations, multiple lexical realizations, and paraphrases.

Introduce unary frames $F(x)$, binary relations $R(x, y)$, compositions such as $G(F(x))$, category-conditioned continuations, exceptions, competing defaults, and novel lexical realizations. Formal notation is for analysis; main training text should not trivially expose the latent symbolic structure.

Generalization splits withhold entity–functor combinations, category members, paraphrases, compositions, longer chains, and category-boundary/exception cases.

6 Measurements

At intermediate residuals decode

$$z_l = W_U \, \mathrm{LN}_f(r_l), \qquad p_l = \mathrm{softmax}(z_l).$$

Measure

$$H_l = -\sum_y p_l(y) \log_2 p_l(y), \qquad \eta_l = \frac{H_l}{\log_2 |V|},$$

and target surprisal

$$S_l(y^*) = -\log_2 p_l(y^*).$$

For a controlled rule-consistent continuation set C, measure

$$P_l(C) = \sum_{y \in C} p_l(y).$$

Capture pre-block, post-attention, and post-MLP states. Do not impose monotonic entropy reduction.

Measure completion performance, target rank/probability, systematic and compositional generalization, substitution invariance, exception handling, residual-update alignment, within/between-category transformation similarity, aligned/orthogonal energy, covariance/effective dimension, head/layer complementarity, and paraphrase invariance.

7 Causal Tests

Use layer/head skipping, activation patching between equivalent functor instances, category-state patching, counterfactual category substitutions, norm-matched random-vector controls, and targeted perturbations. A putative functor is stronger evidence if causal transfer works across compatible substitutions and changes appropriately across incompatible categories.

8 Core Experiments

E1: N-gram to functor. Replace a fixed element of a learned lexical template with members of an equivalence class and test emergence of substitution-invariant transformation.

E2: Learned categories as soft types. After the competence gate, test whether held-out category members elicit similar functor-conditioned trajectories.

E3: Type-sensitive dispatch. Hold syntax fixed while changing learned category.

E4: Novel arguments. Train on $F(a), F(b), F(c)$ and withhold $F(d)$; repeat broadly.

E5: Composition. Test $G(F(x))$, $R(F(x), y)$, and longer chains for sequentially conditioned transformations.

E6: Exceptions. Contradict frequent/default transformations and distinguish early priming from later correction.

9 Controls

Use random and undertrained checkpoints, shuffled category assignments, shuffled functor–continuation mappings, unigram/frequency-matched baselines, lexical identity controls, paraphrases, category-label leakage checks, context-length controls, and multiple seeds. The core factorial comparison is same syntax/different category versus different syntax/same category.

10 Interpretation

Evidence for fuzzy rule-like computation should require convergence of held-out systematic generalization, substitution/paraphrase invariance, learned semantic category dependence, residual/entropy trajectories, and causal transfer/disruption. No attention map, probe, cluster, or decoder alone establishes a symbolic rule.

If supported, the intended conclusion is that continuous transformer computation can develop emergent symbolic-like behavior: approximate objects, semantic types, relations, and reusable transformations arise as functional regularities of state-conditioned residual computation.

11 Relationship to Papers 0.5 and 0.6

Paper 0.5 establishes lexical continuation equivalence, non-monotone uncertainty trajectories, common residual components, and causal equivalent-versus-nonequivalent replacement. Paper 0.6 asks whether semantic categories and hierarchies are learned through text completion, but its current run fails behavioral competence; its attractive geometry is retained as a negative control. Paper 0.7 cannot combine a validated predictive relation with an invalid semantic type system. The next admissible action is to repair and independently validate Paper 0.6 competence under the frozen gate, not to weaken Paper 0.7's entry criterion.

12 Falsification

The hypothesis is weakened if apparent rules disappear on held-out combinations, reduce to lexical frequency, fail category substitution, lack causal transfer, or appear equally in incompetent/undertrained models. A mixed result—local, approximate, context-dependent rule-like behavior—is a valid and informative outcome.

13 Reproducibility and Current Decision

The dependency audit is regenerated with

```
PYTHONPATH=src python -m cl.experiments.paper07_eligibility
```

and writes `results/eligibility_decision.json`, including source hashes, threshold, accuracy range, and the blocked decision. No model weights, selectors, or persistent state are changed.

14 Conclusion

Paper 0.7 is a ready but inactive causal protocol. Its semantic prerequisite fails, so claims about typed functors, compositional rules, or emergent symbolic-like computation would be unsupported. Preserving the stop condition is the result: behavioral competence must precede representation, causal transfer, and rule-level interpretation.